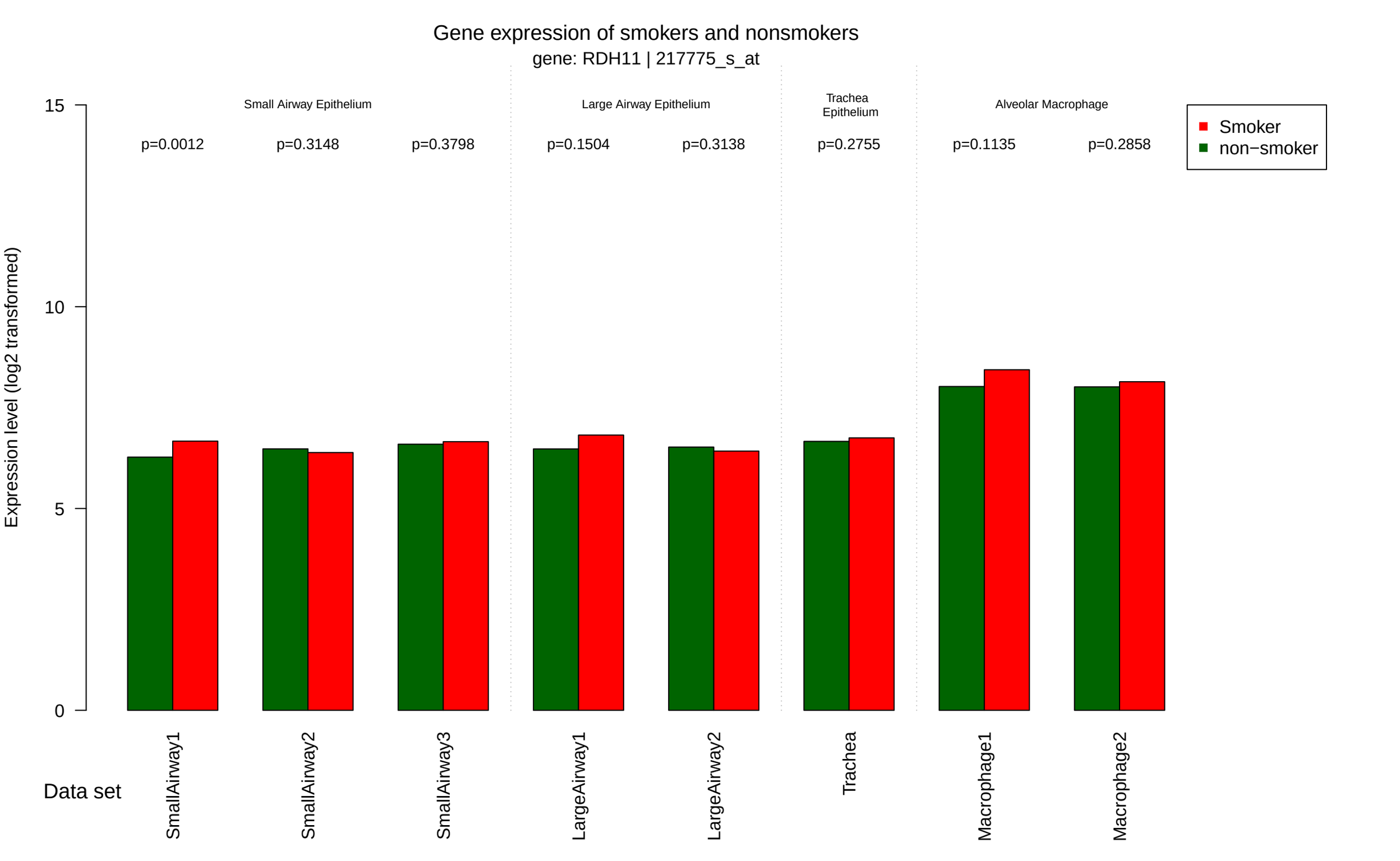
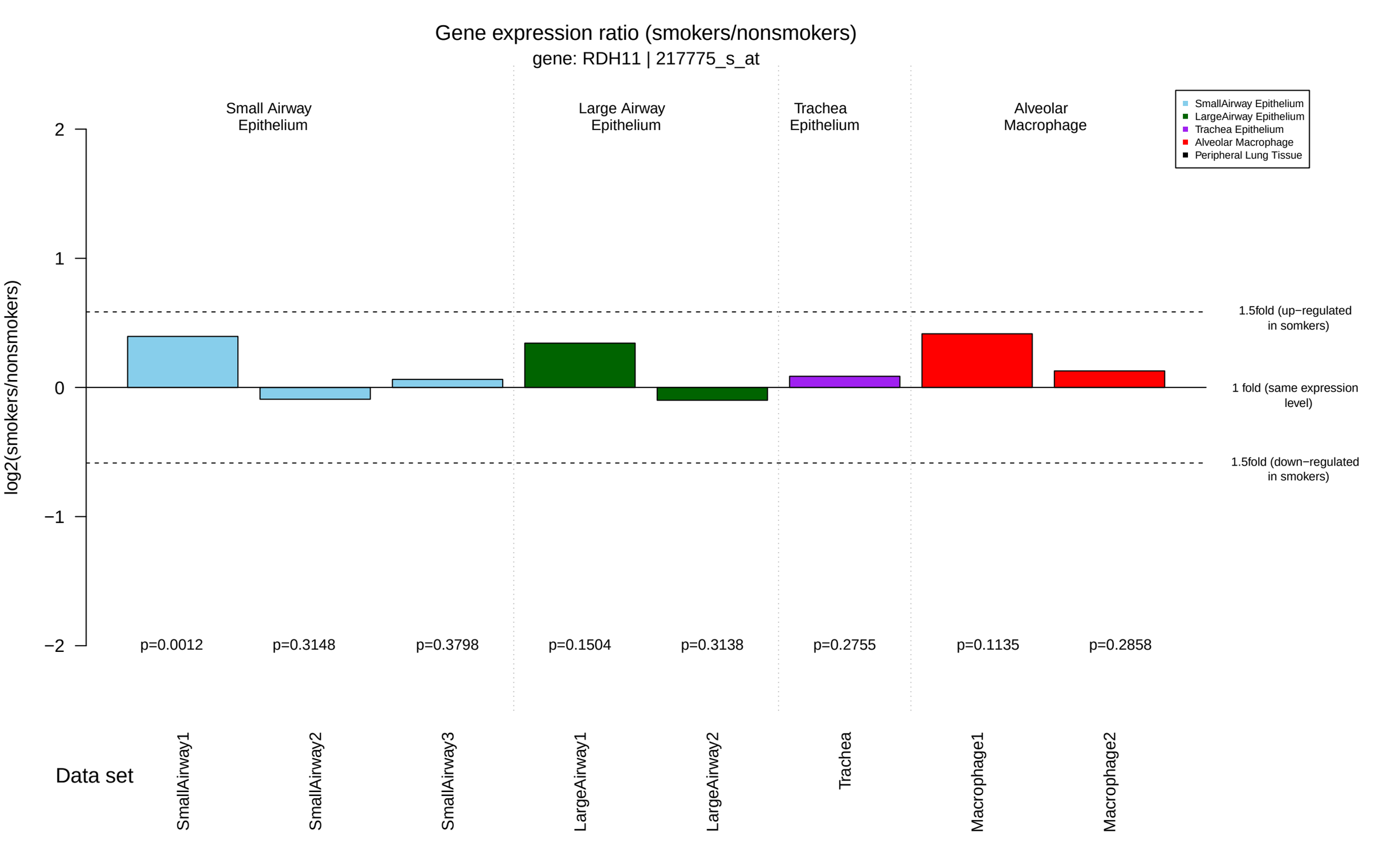
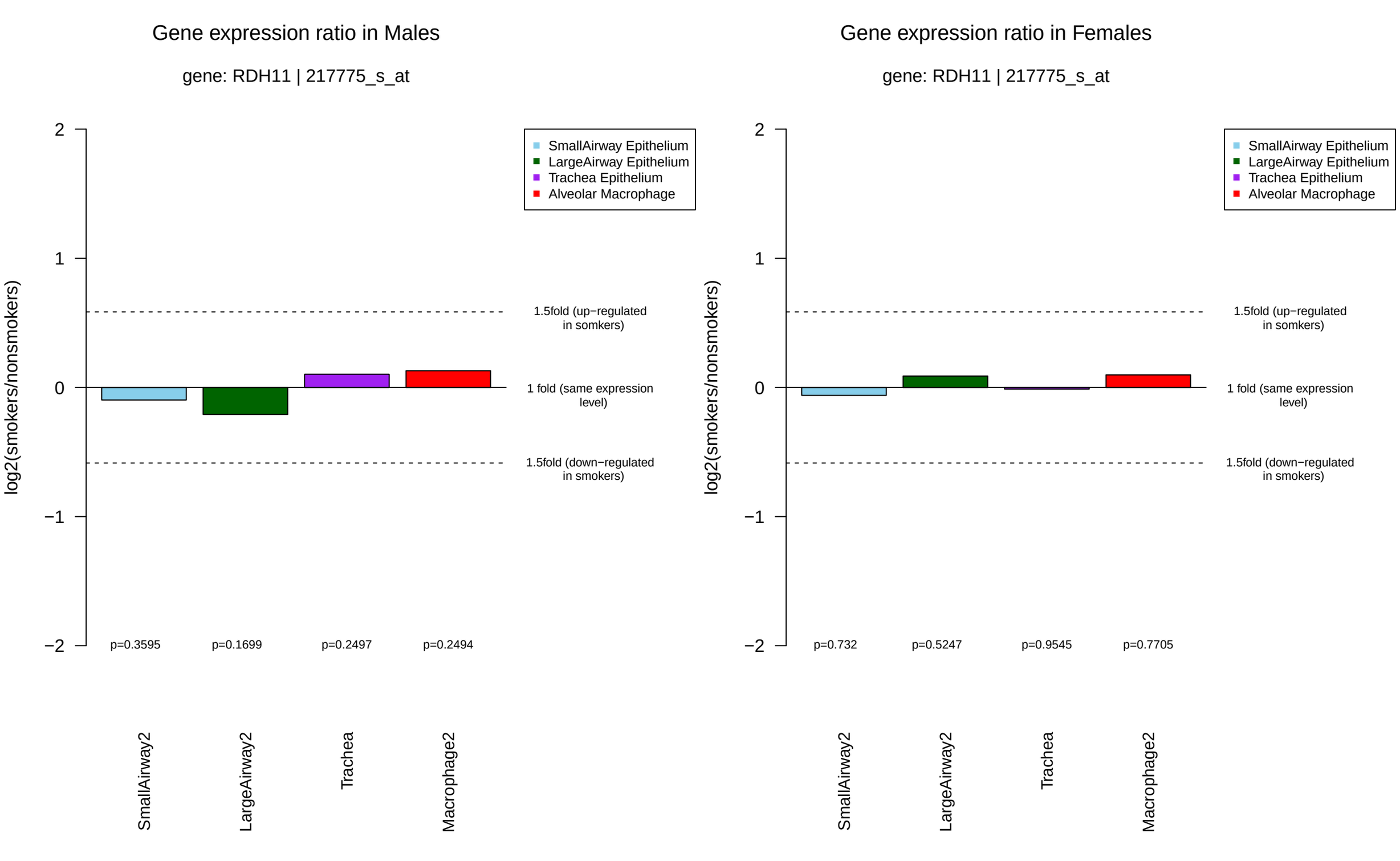
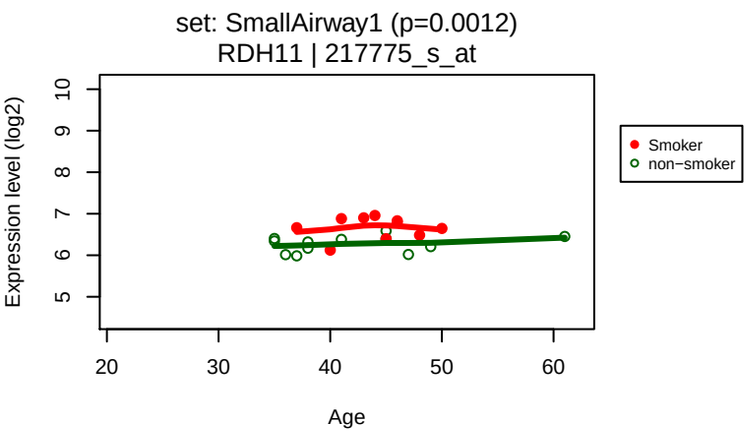
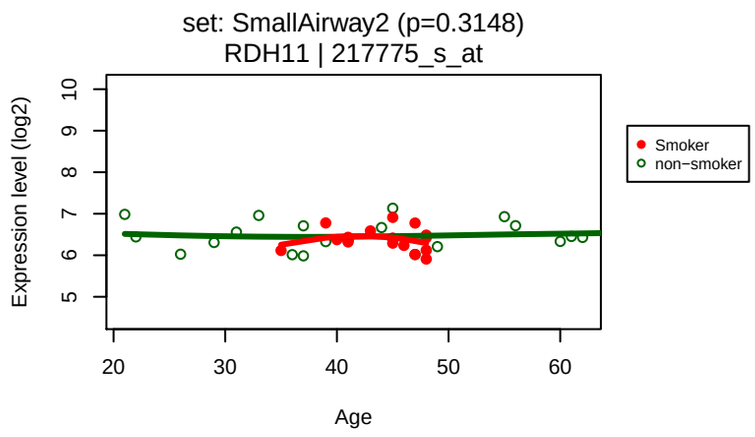
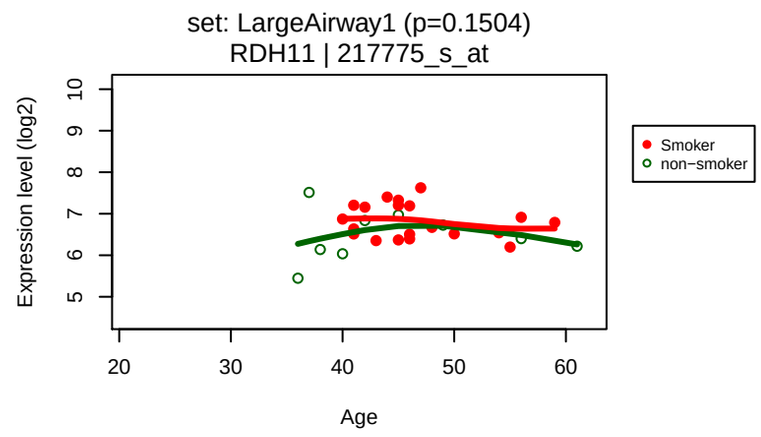

Gene expression of smokers and nonsmokers
gene: RDH11 | 217775_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0012
p=0.3148
p=0.3798
p=0.1504
p=0.3138
p=0.2755
p=0.1135
p=0.2858
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

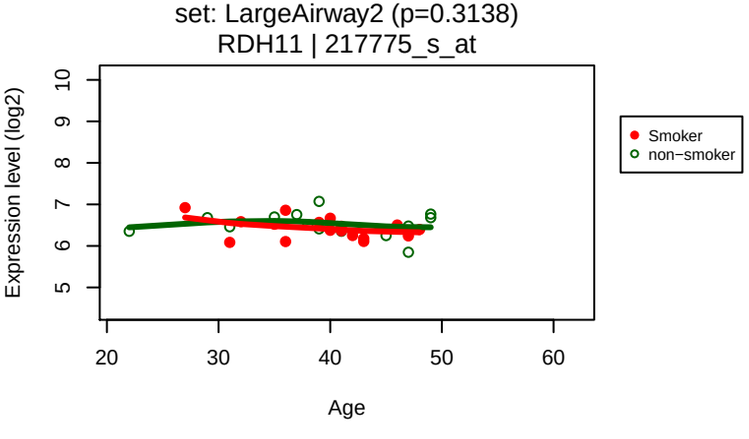

Gene expression ratio (smokers/nonsmokers)
gene: RDH11 | 217775_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.3148
p=0.3798
p=0.1504
p=0.3138
p=0.2755
p=0.1135
p=0.2858
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

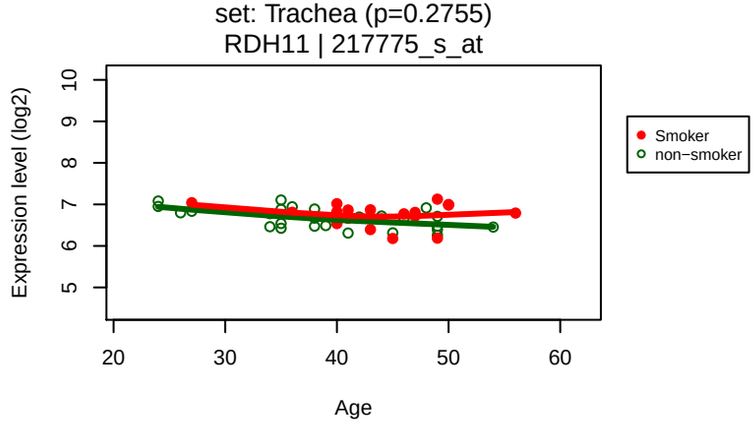

Gene expression ratio in Males
gene: RDH11 | 217775_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3595
p=0.1699
p=0.2497
p=0.2494
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH11 | 217775_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.732
p=0.5247
p=0.9545
p=0.7705
SmallAirway2
LargeAirway2
Trachea
Macrophage2

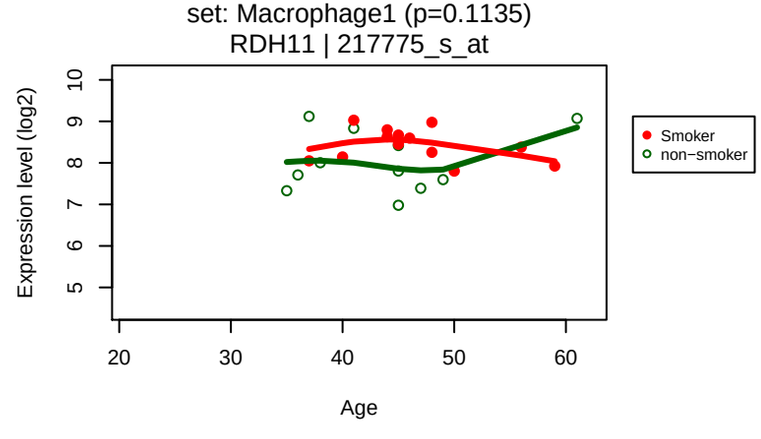

set: SmallAirway1 (p=0.0012)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker

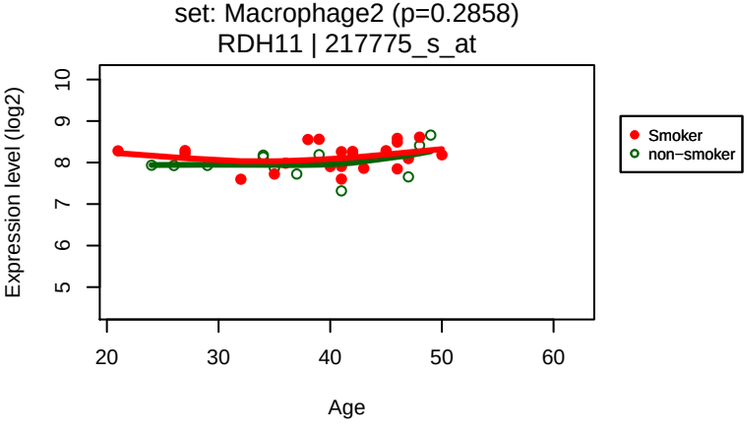

set: SmallAirway2 (p=0.3148)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker

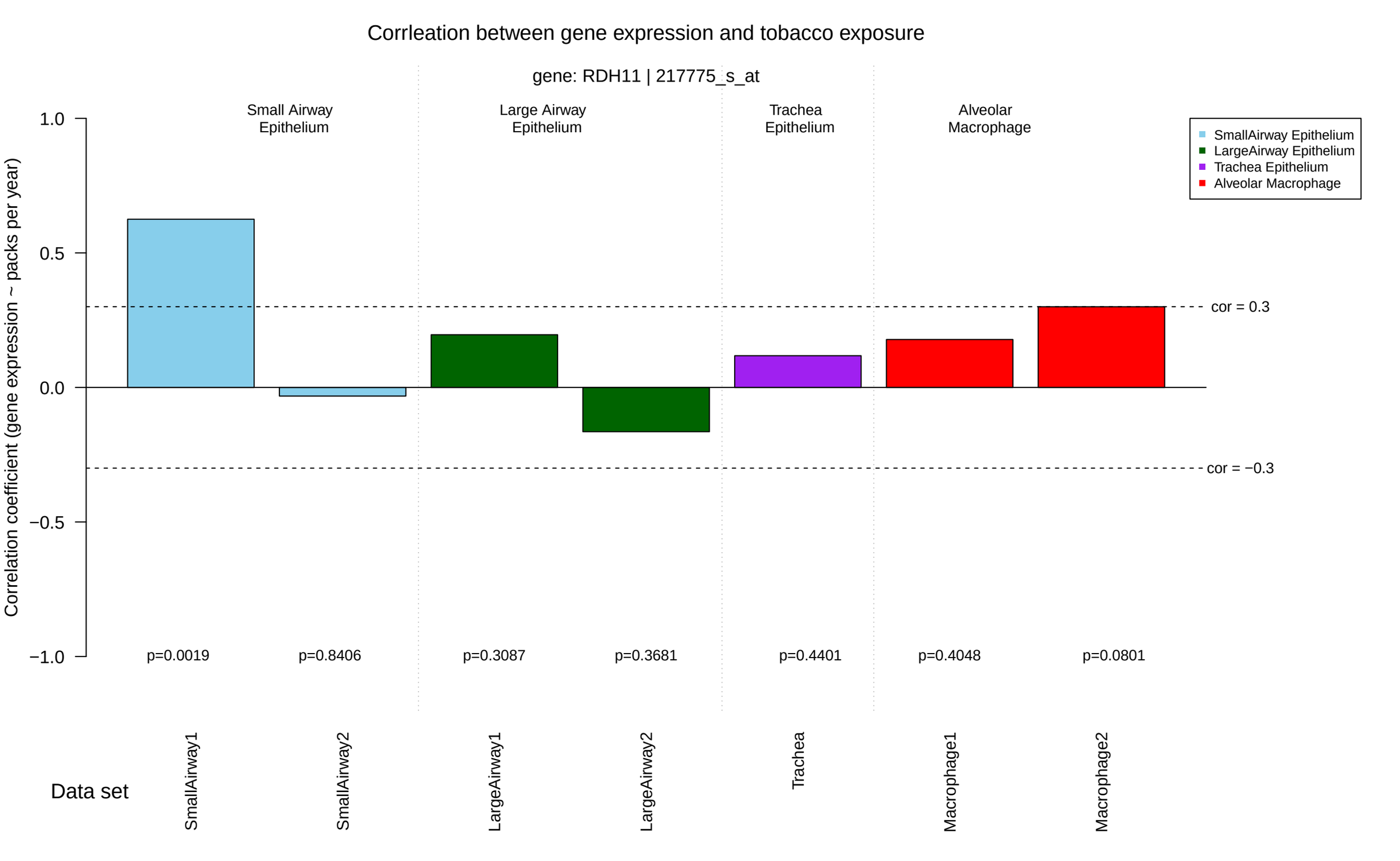

set: LargeAirway1 (p=0.1504)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3138)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2755)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1135)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2858)
RDH11 | 217775_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH11 | 217775_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0019
p=0.8406
p=0.3087
p=0.3681
p=0.4401
p=0.4048
p=0.0801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2